

Experimental Protocol for Studying Pedestrian Behavior in Interaction with Autonomous Vehicles

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Abstract—The field of Autonomous Vehicles (AV) has seen significant advancements in recent years, making the prospect of fully autonomous driving more realistic than ever, largely due to breakthroughs in Machine Learning. A crucial aspect of autonomous driving is the ability to accurately interpret the surrounding environment, as misinterpretations can lead to dangerous situations for other road users. Among them, pedestrians are particularly vulnerable, requiring AVs to develop a precise understanding of their behavior to ensure safety. However, pedestrian behavior is unpredictable and can place them at risk, making AV-pedestrian interactions particularly challenging. To reduce uncertainty in predicting pedestrian actions, AVs can enhance safety by communicating their intended path. Yet, the impact of such communication on pedestrian behavior remains underexplored in current research. Understanding this influence is key to improving pedestrian behavior models, ultimately leading to more effective simulations for training AV control systems. This paper presents a novel experimental study design for examining the effects of different AV-to-pedestrian communication methods on pedestrian behavior.

Index Terms—Autonomous vehicle interaction, VR environment, Pedestrian Behavior, V2P Communication

I. INTRODUCTION

Most accidents are due to human error, but new technologies can reduce the risk of accidents. Advanced Driver Assistance Systems (ADAS) are electronic systems designed to improve safety. First, they detect road hazards and human errors with a set of sensors. Second, they communicate information to the driver or act directly on the controls. ADAS have become increasingly complex, improving the quality and quantity of sensors in recent vehicles. These systems act as early stages and provide technical support for the transition to a fully autonomous self-driving vehicle.

In the dynamic field of autonomous vehicles, the most challenging objective is interaction with other road users. Autonomous vehicles can cooperate with each other with certainty (Connected Vehicles), they adjust their path according to the information given by V2V communication. The communication between AVs shows great success in dealing with intersections and platooning. Facing non-autonomous road users is however much more difficult because of the uncertainty of their trajectory. The pedestrian agent is likely to change its current trajectory because of falls, decision changes,

and other mistakes. The communication from the pedestrian to the AV can also be inconsistent or non-existent, the AV must be as reactive as possible to deal with it. To enhance the understanding of pedestrian agent behavior, this paper focuses on the pedestrian point of view.

In a previous article [1], we showed that the Deep Reinforcement Learning methods are a good solution to deal with pedestrian uncertainty. These methods respond more quickly to the dangerous situations described above, preventing accidents and optimizing the performance of the interaction. To converge, these solutions need to be trained with a large number of scenarios, and we want to create a realistic simulation of the interaction to address this issue. In the simulation, the pedestrian agent must have realistic characteristics to facilitate the transition from simulation to reality.

In addition, to reduce uncertainty, the AV needs to communicate its trajectory with the pedestrian as much as possible. There are many different types of communication, but the light signal and car speed profile (deceleration and acceleration) seem the most promising [2], [3]. We need to test the pedestrian's reactions to ensure that the communication tool enhances the safety and efficiency of the interaction.

For the two reasons mentioned above, we want to conduct a pedestrian test campaign. This campaign will focus on three different objectives: apprehend the pedestrian trajectory around the crosswalk, define the main factors involved in pedestrian decision-making, and analyze the impact of AV communication on pedestrians.

II. RELATED WORK

When it comes to pedestrians interacting with AVs, two key points of pedestrian behavior that require detection by the AV, are the detection of the pedestrian willingness to cross (pedestrian intention) and the pedestrian decision to cross (active choice) [4].

For the first part, studies focus on the analysis of pedestrian behavior. After detecting the pedestrian, the AV needs to classify pedestrians as those who want to cross and those who do not. [2] and [5] aim to define patterns in the pedestrian action plan. They assume that pedestrians communicate their purpose unconsciously or not through non-verbal communication [6]. [7] even asks people how they can predict pedestrian willingness based on partial pedestrian perception. This article

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shows how the position of body parts is worth considering factors for classification.

For the second part, AV needs to analyze and predict pedestrian decisions to prevent accidents. To this end, many studies try to identify, list, and order the factors involved in the decision-making. [8]–[10] state that pedestrians have a personal safety margin: when there is enough time to cross, pedestrians will cross. Tests were carried out on numerous crosswalks and data was collected using static cameras [11]. The results show that many factors impact the safety margin: pedestrian features (age, sex, stress, etc.), car features (speed profile, traffic flow, vehicle size, etc.) and environmental features (road size, lane number, weather condition, other pedestrians, etc.). Another study [12] considers the trust of the pedestrian: if the pedestrian believes that the car will yield, then he/she will cross. To evaluate trust, they provide the subject with a press-button and instruct them to press it when they feel safe to cross. The results reveal that pedestrian's risk assessment plays a major role in the decision-making. The crossing environment and the pedestrian's ability to perceive it therefore have a significant impact on his/her behavior [13].

After detecting and analyzing the pedestrian's behavior in order to react to it, V2P (vehicle-to-pedestrian) communication can be used to influence the pedestrian's behavior, thus improving the interaction safety and efficiency. Indeed, article [14] shows that pedestrians are often looking for the driver's attention for crossing. Traditionally, detection signs such as direct eye contact or strong deceleration indicate the driver's awareness to pedestrians. Moreover, [15] shows the importance of driver behavior to increase the confidence level of pedestrians.

To address this issue, V2P communication is integrated using a Human-Machine Interface (HMI) system. The variety of HMI for V2P communication is vast, and this subject remains an active area of study. The communication can be carried by a light signal on the car or displayed on the road [15]–[21], induced by the speed profile [15], [20], sent using a cellphone message [22], etc. Many studies focus on communication through light signals using LED light strips of different colors [2], [16], [17], [19], [20].

Four criteria can be considered in the design of an effective HMI for V2P communication: the light color, the signal position [16], the type of signal (such as a line, a symbol or a text), and the message content (information or advice). Regarding the color selection, some studies pick red and green [16], [20] while others use the cyan color [17], [19]. Those colors are intuitive because pedestrians are familiar with the road signalization. For the signal position, there is no consensus on the subject. [16] conducts a study to analyze the pedestrian's gaze on the environment during the interaction. With this information, we could place the light signal at an ideal position for it to be seen. In [16], pedestrians' gaze is primarily focused on the bumper and the road. In another article [3], the windscreen display was more efficient for recognizability. Comparing the positions of the HMI is

challenging, as the pedestrian's focus varies depending on the distance between the two agents [16] and the communication effectiveness can be compared by several points [3] such as recognizability (identify the signal), intuitive comprehensibility (understand the signal), and unambiguousness (non-contradictory messages). Concerning the type of signal, [3] shows that the LED light strip is not very effective in terms of recognition, whereas the best type of message is the symbol type. Moreover, according to the same article, the message content given by the car must be an indication of the vehicle's desired actions for the pedestrian, such as traffic lights, and not about the vehicle information, to reduce ambiguity.

Article [18], however, reveals some limitations of the HMI system such as pedestrian collaboration: the car needs the pedestrian to act as planned and distractions, such as other pedestrians or cell phones, may alter their ability to do so. Therefore, HMI is an efficient tool to replace the presence of the driver during an interaction, but not sufficient to ensure safety. In all cases, defining a standardized mode of communication is crucial for the comfort of pedestrians and for building a habit of interaction with AVs [15].

We want to design an experiment that allows us to analyze pedestrian behavior during an interaction with an autonomous vehicle integrating V2P communications. To design it, we analyze different pedestrian campaigns and VR campaigns especially. As summarized in the table I, these campaigns can be categorized into four types: **the survey campaign, the 2D screen tests, the real tests and the VR tests.**

TABLE I
STUDY TYPES AND THEIR CHARACTERISTICS

Type of Study	Advantages	Disadvantages
Survey	Sample size Material support	Very limited realism Data quality
2D screen	Visual perception	Limited realism
Real-world	Environment immersion	Participant Safety Data collection Replicability
Virtual reality	Good immersion Participant Safety Data collection Replicability	Invulnerability effect

The first and second types of study focus on the pedestrian perception of his/her environment. Pedestrian data can be collected through surveys, offering the advantage of a large sample size and lower technical requirements compared to other campaigns. Types of studies are often combined, as seen in [17], where the authors use pictures in a survey about LED light strip-based V2P communication. The subject expresses his/her impression on colors, and animations at different positions. [3] shows participants 20 videos of vehicles equipped with different HMIs and speed profiles to measure their effectiveness in front of a pedestrian. This study shows that a projection gives better results, but the sun conditions can alter the performances of such communication. The good solution then is to place the display in the most visible place in the vehicle. [23] uses the same idea to compare different

HMI. Despite the lack of realism of such an experimental setup, such a study can serve as an introduction to targeting promising HMIs.

The third type of study uses real tests (pedestrians facing a real car) with extensive technical support. [16] uses a mechanical device to measure the willingness of each participant to cross and collect their gaze in realtime. The surprise effect is not taken into account, as information on different HMIs is provided in advance to the participants. [12] creates an Intent Communication System (ICS) and tests it in a real experiment. A survey is carried out to extract pedestrian feedback. Merging the survey and the actual tests produces more comprehensive and exhaustive results.

Finally, the fourth type of study uses virtual reality (VR) to ensure pedestrian safety while testing a real-life AV model confronted with a pedestrian. This method looks promising because it is close to reality, allows pedestrian movement during testing (wireless), requires less technical support and data extraction can be done directly in the VR headset. Moreover, the pedestrian behavior in the VR experiment is realistic according to [24].

Several limitations are worth mentioning in the current literature. To address these, propose several improvements in the design of the experimental study of pedestrian behavior facing AV.

First, the article [15] highlights a significant gap in studies concerning the application of multi-modal solutions, which integrate various communication tools for pedestrians. Our research aims to analyze the impact of integrating both explicit and implicit communication on pedestrian behavior, as this combination needs to be further investigated. Additionally, while many studies only focus on the use of lightstrip tool, we intend to explore and apply the best available tools from the current state of the art.

Second, many studies [19], [20], [23] rely on pressing tools to indicate pedestrian intent to cross. However, this method introduces biases compared to real-world crossing, as demonstrated by [14]. Even studies that let the participant cross are limited by physical constraints, such as cables, which may introduce additional biases [18]. We propose a study based on the use of a wireless VR headset that allows the participant to move freely in the interaction space.

Third, the impact of pedestrian habituation to autonomous vehicles, particularly with regard to integrating new communication tools, remains largely unexplored. Understanding the pedestrian learning process in this context is crucial, and we believe further research is needed to fully comprehend how pedestrians adapt to these technologies over time.

III. EXPERIMENTAL SETUP

We aim to conduct an experimental campaign using VR. This requires building a test environment, defining the study procedure, and identifying key factors for analyzing pedestrian behavior.

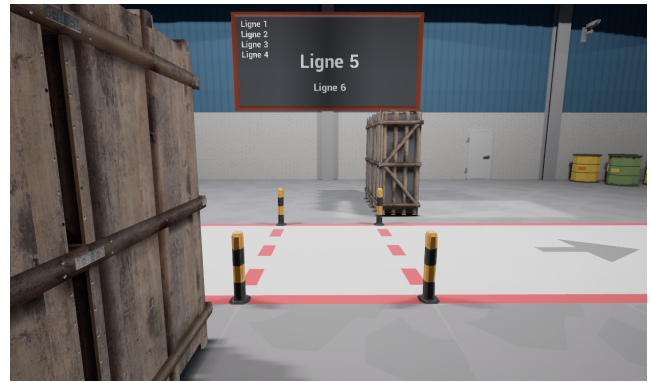


Fig. 1. The virtual interaction environment: the vehicle-pedestrian interaction zone is delimited by the dotted lines marks

A. Test Environment

Our experiment takes place in a closed industrial environment for two main reasons. First, this environment is more subject to the integration of autonomous vehicles coexisting with workers. Second, it provides a controlled environment, well-suited for AV-pedestrian interactions.

To ensure the realism of pedestrian behavior, we want the environment to be as intuitive as possible. The experiment must balance the realism of simulation and the simplicity of experience: the more realism we add, the more the pedestrian will be distracted by those details. In the other case, if the simulation is too simplified, the pedestrian may not act as he/she would normally do. There are solutions to simplify the experience while adding realism: minimizing distractions while immersing the participant in the environment, incorporating elements to guide the participant, and preventing inconsistencies (unrealistic elements).

The first idea is to limit the number of objects as it may significantly impact the pedestrian analysis time. The only objects used are highly representative of the industrial environment to enhance the participant's immersion. To this end, the vehicles are forklift trucks, industrial noises can be heard, and the participants are placed in a big warehouse with some shelves loaded with boxes.

The second idea is implementing helping elements to facilitate pedestrian comprehension. The board on the walls indicates basic information about the scenario. The pedestrian crosswalk is marked with a colored square on the road. It informs the pedestrian of the danger area. The road is one-way and arrows are pointing in the correct direction, to influence the pedestrian to look to the left. A luminous halo hovers over the area where the pedestrian is supposed to go. These aids must be sufficiently intuitive for the pedestrian to understand their meaning on their own. The participant must be informed previously about these to avoid a surprise effect that could alter the results of the experiment.

We also placed some occlusive shelves at the left of the pedestrian to prevent the participant from seeing the arriving car before the start of the scenario. This element makes the experiment more realistic compared with other options such

as appearing vehicles and vehicles stopped before the start of the scenario. The lack of information about the left part of the warehouse also increases the perceived danger from crossing for pedestrians. While some studies use vehicles coming from adjacent streets, we considered this option unsuitable for interactions that occurred quickly due to the pedestrian's lack of perception of the vehicle's proximity.

B. Scenario Procedure

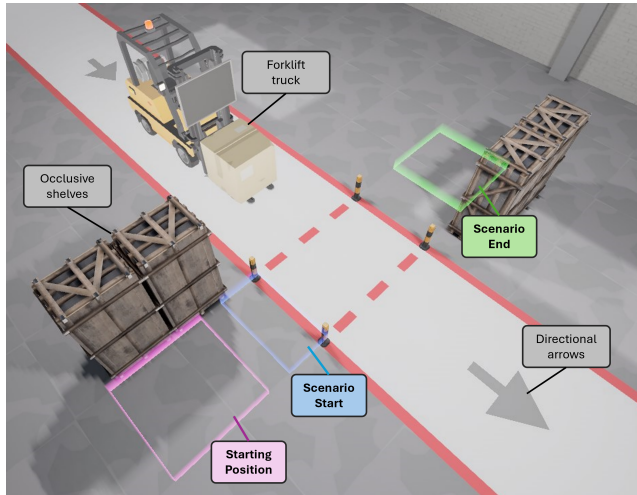


Fig. 2. The different areas and their roles in the scenario procedure

The starting position of the participants is in a purple area located far from the crosswalk (first crossing, Fig.2). When the participant is ready, he/she pushes a button, starting the scenarios. The participant actively chooses when to start the experience. This is important to make sure that the participant is ready, which a third-party start or timed start cannot do. At this step, the purple area turns off and a new area lights up in blue near the crosswalk.

The scenario starts when the pedestrian enters the blue area. At this moment, vehicles start to drive and the participant can interact with a specific traffic flow. The scenario continues until the participant crosses the street and enters the arrival area represented by a green area.

When the participant enters, the scenario is stopped and a new one is displayed without any interruption because both sides of the road are similar. Indeed, the environment is identical in one way and the other. This fact increases the realism of the experiment because the participant is not teleported or his/her headset is removed between the tests. The green area disappears and we loop on the appearance of the blue zone for starting the new scenario.

The participants will conduct a series of crossings. Between those tests, the participants hear a countdown of 3 seconds to let them turn around (also displayed on the board). Once the countdown ends, the board signals them to cross the road again. At the end of the experiment, the board requests the arrival of a new participant.

C. V2P Communication

In our experiment, we design the eHMI based on the literature, as described in the state of the art. We suggest that good V2P communication relies on two parts: a display from the car and an adjusted speed profile for pedestrian detection (whether the car is changing its trajectory).

To evaluate their effectiveness, we analyze the pedestrian reactions facing 4 different scenarios: no HMI, adapted AV speed profile, display on the vehicle, and both communication tools. Based on the literature, we will display an animated and colored symbol on the windscreen of the vehicle with indications for pedestrians as illustrated in Fig. 3.

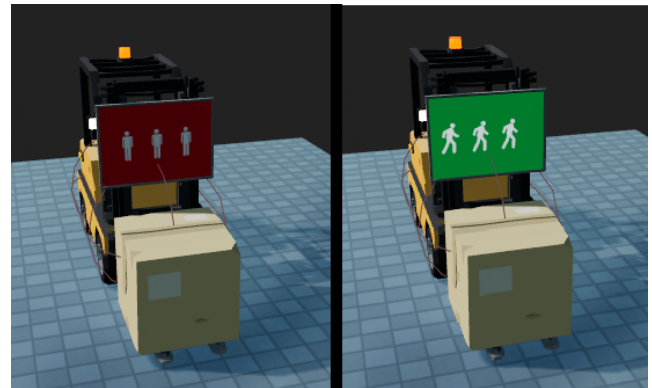


Fig. 3. Vehicle to pedestrian communication in our virtual environment (left: wait signal, right: cross signal)

Our communication system has the following features:

- Colored red and green: prove to be well-recognized
- Animated: attract pedestrian attention
- On the windshield: Visible part of the vehicle
- Symbol message: improve the interaction comfort
- Advice for pedestrian: reducing message ambiguity

One of the most important criteria is to minimize message ambiguity. We figure that the best possible symbol is the same two already used for the pedestrian crosswalk signalization. Indeed, pedestrians are more accustomed to receiving messages that advise them on their actions, rather than being informed about the vehicle's actions [3]. This choice may also lead to a reduction in adaptation time, as pedestrians are used to this method of communication.

Another important factor is the moment we display information to the pedestrian. In this experiment, the AV displays its choice when deciding on the crossing order. If the pedestrian is not acting according to the AV plan: the pedestrian crosses even if a red signal is displayed, the AV must react. It will not change its signal but apply if it can an emergency brake and it will honk. It's a proportionate reaction between an AV which drives with no regard for the pedestrian who puts himself in danger and an AV which displays a green signal even though it is unable to stop. In the first case, the pedestrian is put in danger and can not interact with the AV, in the other case, the pedestrian always has the right of way and may put himself in danger. It is crucial to establish a trusting attitude between the pedestrian and the vehicle communication system.

D. Scenarios Sequence

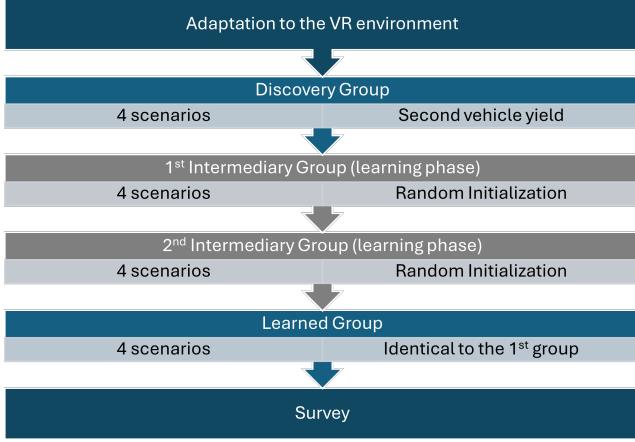


Fig. 4. Study procedure for a participant

As shown in Fig. 4, we want the pedestrian to get used to the VR environment during the first phase of the study procedure. The participant is adapting to the VR technology and the warehouse environment by letting him/her freely walk. This is crucial because otherwise the analysis will be biased by the adaptation to the environment, as highlighted in article [20]. After this adaptation period, we can launch the scenarios with the participant's agreement.

The experiment consists of four scenario groups. The first group is the discovery stage for the different methods of communication. The second and third groups are training groups, where the participants can adjust their behavior accordingly and learn from their mistakes. The fourth and last group is the same as the first one. In this group, the pedestrian is considered to be used to interact with the AV. This similarity between the 1st and the 4th group eliminates the influence of situational differences (environmental factors) on participants' crossings. As a result, it allows us to analyze changes in pedestrians' behavior through the learning process, following his/her interactions with different communication tools. This choice is facilitated by the replicability of experiences, which is made possible through the use of virtual reality (VR).

The main reason for creating intermediary groups is to prevent bias caused by pedestrians getting used to a fixed experience. Indeed, the first and fourth groups have the same initialization features to enable comparison of results. The initialization of the two intermediary groups, however, is random in many ways: number of follow cars, initial positions of the cars, and position of the car choosing to yield. It is important to state that the scenarios in each group are randomly ordered for the same reason: to avoid the pedestrian from getting used to a specific order. Preventing expectations is crucial for the quality of the results.

IV. PEDESTRIAN BEHAVIOR ANALYSIS

In this study, we propose an analytic methodology to understand the impact of the AV presence and its V2P communication on pedestrian behavior. After designing a realistic

interaction scenario between the participant evolving in the environment as a pedestrian, and an autonomous vehicle, we then need to define relevant data and the solutions to extract them from this study.

TABLE II
FEATURES RELATED TO THE PEDESTRIAN AND VEHICLE INTERACTION
EXTRACTED USING THE VR HEADSET.

Features	Description	Units
Elapsed time	Time elapsed	s
Ped. p	Pedestrian position	m
Ped. v	Pedestrian's instantaneous velocity	m/s
Ped. a	Pedestrian angle vision	deg
Ped. in DZ	Pedestrian is in the danger zone	Ø
Distance in DZ	Distance traveled by the pedestrian in the danger zone	m
Time in DZ	Time spent by the ped. in danger zone	s
Avg Ped. v in DZ	Average ped. velocity in danger zone	m/s
Ped. estimated exit time	Estimated time for the pedestrian to exit the danger zone	s
Veh.1 p	Position of vehicle 1	m
Veh.1 v	Velocity of vehicle 1	m/s
Veh.1 a	Acceleration of vehicle 1	m/s ²
Veh.1 passed	Vehicle 1 passed through the crosswalk	Ø
Veh.1 Signal	Signal of vehicle 1	Ø
Veh.2 x	Position of vehicle 2	m
Veh.2 v	Velocity of vehicle 2	m/s
Veh.2 a	Acceleration of vehicle 2	m/s ²
Veh.2 passed	Vehicle 2 passed through the crosswalk	Ø
Veh. 2 Signal	Signal of vehicle 2	Ø

A. Data Selection

We will focus on three groups of data: factors influencing pedestrian behavior, characteristics of the crossing, and the criteria for successful interaction.

First, pedestrian behavior is influenced by many factors, both personal and environmental [15]. It varies according to several individual factors: demographic data such as age, gender, and type of familiar environment (VR, urban, rural, industrial); and perceptual data such as confidence, stress level, and scenario preferences.

Second, the AV-pedestrian interaction is examined from both physical and emotional perspectives. On the one hand, real-time data on the vehicles and the pedestrian are collected, such as their odometric data during the crossing, as well as temporal information such as waiting time, crossing duration, and arrival time. These elements could be used to evaluate key pedestrian features, such as his/her decision to cross (critical gap). On the other hand, participant perception of the interaction is gathered for each scenario type.

Third, we want to extract features to evaluate the quality of the interaction in terms of safety and temporal efficiency. For safety, we consider metrics such as safety margin (time gap between pedestrian exit and vehicle arrival) and collision risk (AV ability to stop before the crosswalk). For temporal efficiency, we evaluate the lost time for the vehicle and the pedestrian during the interaction.

Results can also be compared with other experiments. In particular, we aim to analyze the impact of VR use on pedestrian behavior. One key characteristic of this analysis is the average pedestrian velocity. Indeed, there is no consensus on the impact of VR use on pedestrian speed: [24] states that

the velocity difference is not significant, whereas other studies consider a small bias. Therefore, we consider comparing our results with the average speed obtained from the literature.

TABLE III
SURVEY QUESTIONS

Type of data	Questions
Scenario Preferences	In which scenario did you feel most at ease? (1 / 2 / 3 / 4)
Stress Level	How stressful did you find the "signaling + adaptive speed" scenario? (from 0 to 4)
Stress Level	How stressful did you find the "adaptive speed" scenario? (from 0 to 4)
Stress Level	How stressful did you find the "signaling" scenario? (from 0 to 4)
Stress Level	How stressful did you find the "no communication" scenario? (from 0 to 4)
V2P Communication	What did you base your crossing decision on? (Signaling, AV Speed, AV Deceleration)
V2P Communication	Did the signaling help you make the decision to cross? (Yes / No)

B. Data Collection

We adopt two methods for collecting data from the experiment. First, the head-mounted display collects data in real-time allowing us to get a better understanding of the pedestrian behavior during a scenario. Information presented in Table II will be collected and stored through three backup solutions: a local backup (on the VR headset), a backup on the laboratory server, and a backup on a local computer. Second, the participant perceptions are gathered through a survey conducted after the test series. This provides valuable insights into the participant's perceptions in terms of safety, confidence to cross the road, and communication effectiveness as presented in Table III.

V. CONCLUSION

In this paper, we design an experimental study to analyze pedestrian trajectory, pedestrian decision-making and effects of direct (light signal) and indirect (speed profile) V2P communication during interactions between a pedestrian and an AV. Our solution implies the use of a VR headset for three main reasons. First, the VR environment provides a realistic experience and ensures the consistent reproduction of the same scenario for all participants (comparison between the participants). Second, the wearable VR headset is minimally intrusive, enabling the participant to move freely as a pedestrian during the interaction (data quality). Third, the use of the VR headset enables quick set-up and real-time data collection.

Following the development of this pedestrian behavioral study using Unreal Engine software, we will conduct the study, in collaboration with the CHU of Besançon, with a sample of around one hundred participants. With the results of this study, we aim to obtain a better understanding of pedestrian behavior, enabling the development of a realistic pedestrian model in terms of movement, decision-making, and adaptation to V2P communications. Preliminary results already show that indirect communication has a greater impact on pedestrian behavior, while the communicative tool increases pedestrians' confidence during the interaction.

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